# A REPORT ON

# TEN-YEAR CREDIT PORTFOLIO MIGRATION, EXPECTED LOSS ESTIMATION, AND STRESS TEST ASSESSMENT

By

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Course Code: FIN F414 Course

## Title: Financial Risk Analytics and Management

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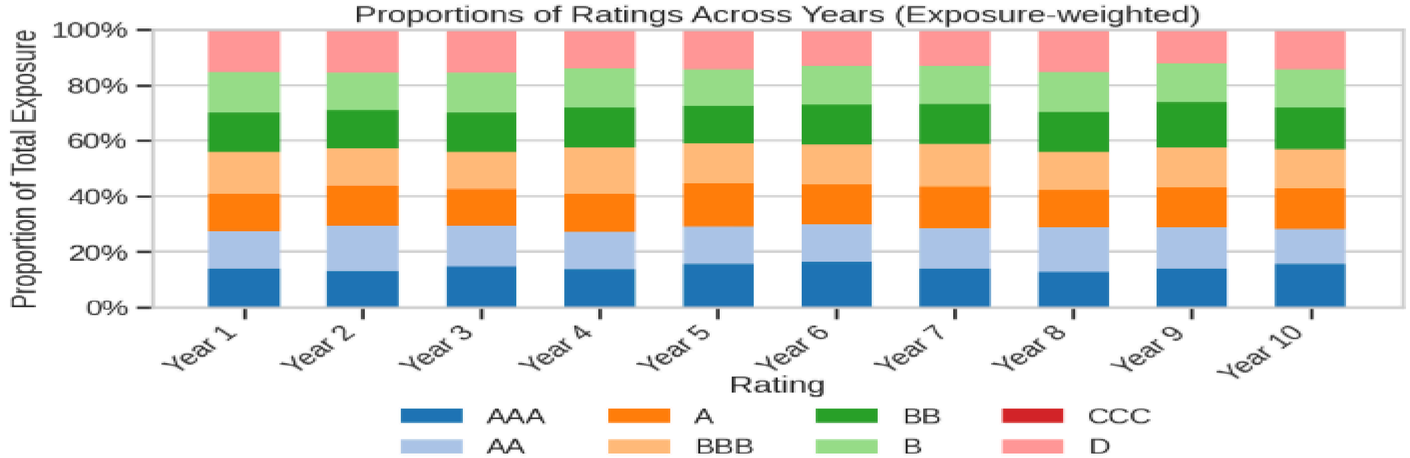
# BANK PORTFOLIO OVERVIEW

## 1. Distribution of Ratings across the years

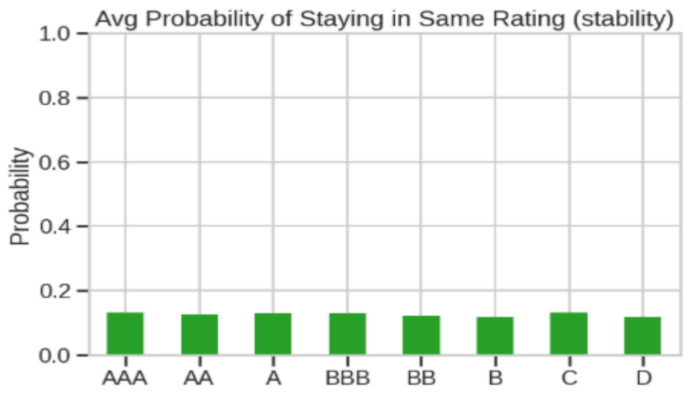
### 1.1. Composition

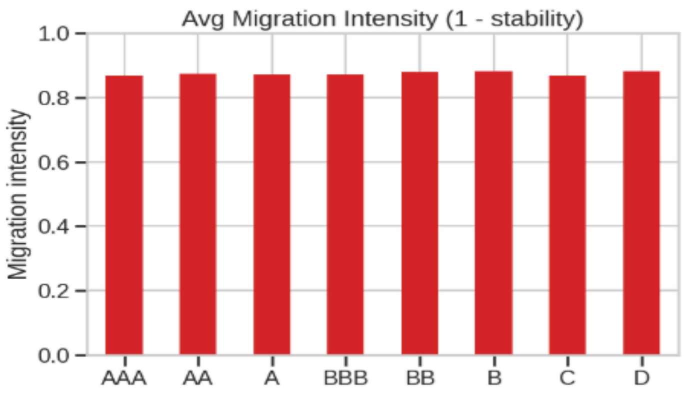
Throughout the ten years, the bank's 2,000 borrowers span every rating category from AAA to D, with exposure fairly evenly spread across the rating spectrum. No single rating class dominates, and each year shows a balanced mix of high, mid-, and low-rated borrowers.



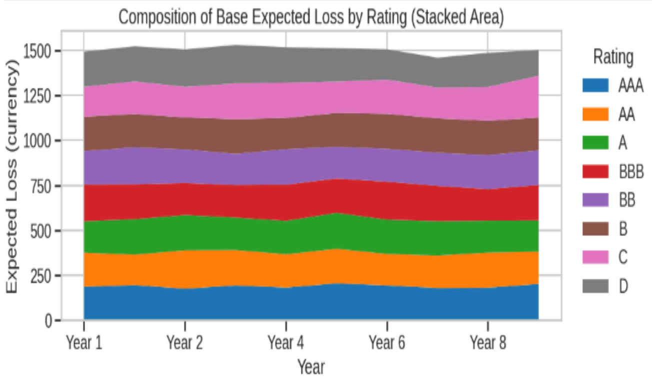

### 1.2. How Often Ratings Stay or Change

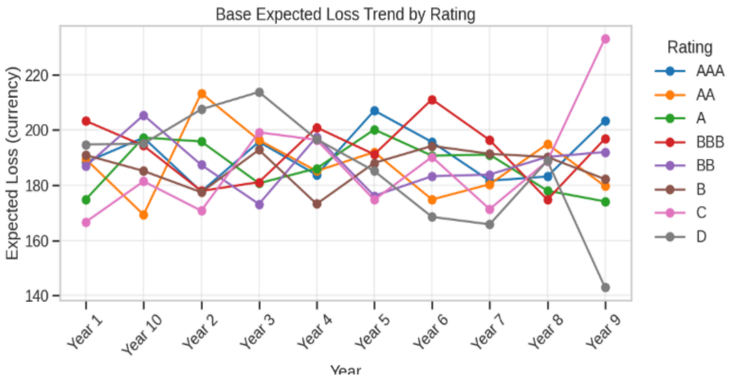
Since all rating groups are characterized by low stability and high movement, the ratings of the portfolio tend to change often rather than stay in the same category.



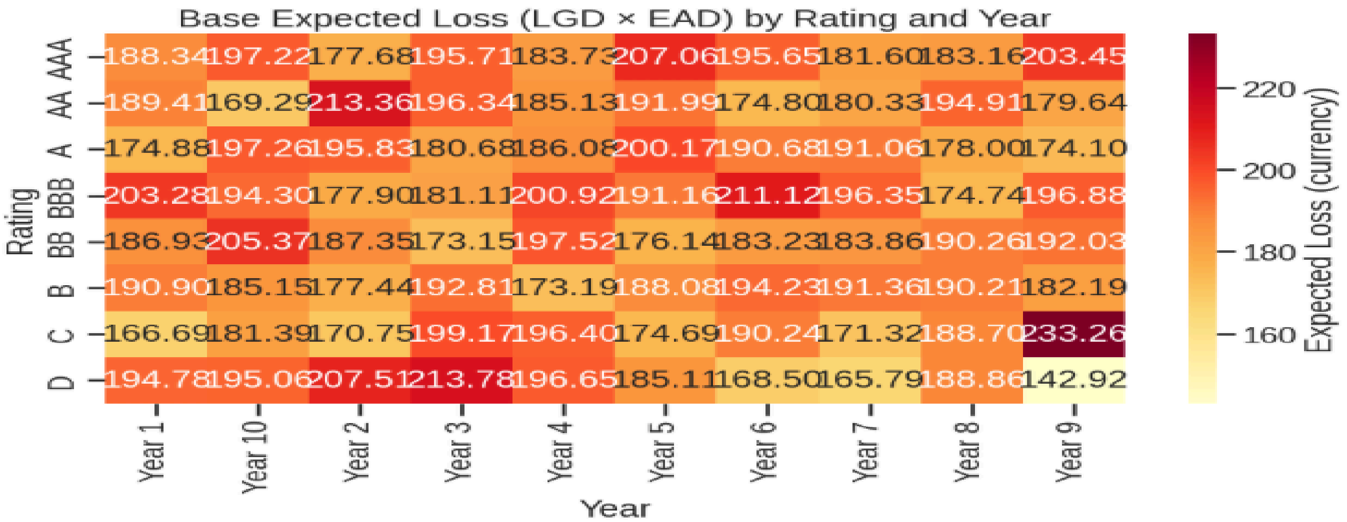




### 1.3. Base Expected Loss Across The Credit Ratings Over The Years

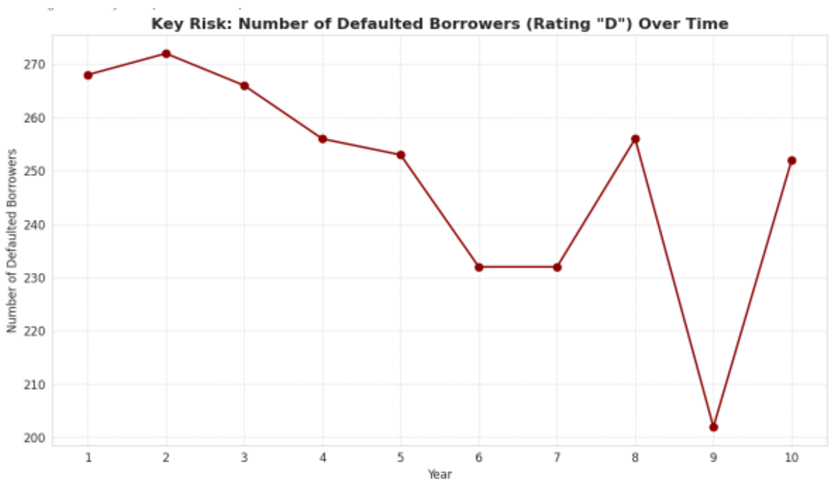


The heatmap shows that the base expected loss is largely similar across all rating categories and years, and values cluster within a relatively narrow band regardless of the credit grade. No rating class has consistently higher or lower EL, which suggests that the differences are primarily driven by EAD and LGD changes as opposed to the rating grade. Year-on-year changes are small and evenly distributed, suggesting a relative stability of base loss across the portfolio and no concentration in any particular rating bucket.




## 1.4. Trend in Defaulted Borrowers (Rating "D") Over Time




Interpretation:

The defaults gradually decrease from Years 1–6, indicating that the portfolio performance is improving. The spike in Year 8 may point to temporary increases in stress and further sharply decreased in Year 9, likely due to recoveries or portfolio cleanup. Year 10 stabilizes closer to long-term average levels.

## 1.5. Overall Distribution Profile of Credit Ratings and Associated Exposures

Distribution of EAD and Base Expected Loss across Rating Grades The above analysis indicates that across the portfolio, the proportionate distributions of EAD and base expected loss are well-balanced, and no single rating class has disproportionately high exposure or high levels of loss. Strong similarities of medians and ranges across categories suggest that risk is not concentrated in particular segments. Exposure and loss drivers are uniform across the rating spectrum, thus showing a diversified credit book in which movements in expected loss are more due to portfolio-wide dynamics rather than structural rating-specific factors.

## 2. Expected Loss And Cumulative Probability of the Bank Portfolio

### 2.1. Expected Loss Estimation (Years 2–10) – Methodology

The approach to estimating Expected Loss for Years 2 to 10 begins with the conversion of year-on-year rating transitions into annual default probabilities. For every pair of consecutive years, a transition matrix is constructed showing how borrowers move across rating categories. The probability of moving from any rating into the default bucket in a given year is taken as the annual PD for that rating.

| | Year 2 | Year 3 | Year 4 | Year 5 | Year 6 | Year 7 | Year 8 | Year 9 | Year 10 |
|---|---|---|---|---|---|---|---|---|---|
| **AAA** | 12.6531 | 8.6207 | 10.9375 | 12.9167 | 10.4478 | 11.3139 | 11.1554 | 8.5714 | 9.9206 |
| **AA** | 15.4812 | 12.5899 | 10.2767 | 13.4454 | 11.3725 | 12.5541 | 14.2857 | 10.4478 | 14.5833 |
| **A** | 13.8655 | 13.3080 | 12.2363 | 11.8852 | 10.9434 | 11.6667 | 8.9219 | 13.6364 | 13.0612 |
| **BBB** | 13.4831 | 16.1157 | 17.6724 | 15.0735 | 9.1270 | 11.0687 | 11.4068 | 7.3276 | 11.1554 |
| **BB** | 12.6984 | 11.6466 | 17.0124 | 12.7490 | 13.5371 | 16.0156 | 12.6984 | 10.7570 | 13.1783 |
| **B** | 12.5475 | 14.1079 | 14.6154 | 10.8333 | 12.0968 | 9.3385 | 16.4062 | 10.9804 | 17.1315 |
| **C** | 13.5965 | 16.1435 | 10.9804 | 13.1274 | 11.3043 | 10.0806 | 13.3621 | 13.1474 | 10.2990 |
| **D** | 14.5522 | 13.9706 | 9.3985 | 10.9375 | 14.2292 | 10.7759 | 14.6552 | 5.8594 | 11.8812 |
| **Portfolio PD** | 13.6041 | 13.3192 | 12.8040 | 12.6768 | 11.5953 | 11.6122 | 12.7833 | 10.0899 | 12.6232 |

Interpretation:

- PDs vary across all years and ratings, showing that default probabilities are not static and adjust annually regardless of initial rating strength.

Once annual PDs are derived, each borrower is assigned the PD of their starting-year rating, meaning all borrowers in the same rating share the same PD for that year. The rating-mapped PDs are used in concert with each borrower's LGD and EAD to calculate Expected Loss first on a borrower-by-borrower basis, and then on both a rating and portfolio-wide basis using:

$$\text{EXPECTED LOSS} = \text{PD} \times \text{LGD} \times \text{EAD}$$

The output presents the expected loss for each credit rating from Year 2 to Year 10, along with the overall portfolio expected loss trend across the same period.

=== Expected Loss (EL) by Credit Rating – each year (₹ crore) ===

| | Year 2 | Year 3 | Year 4 | Year 5 | Year 6 | Year 7 | Year 8 | Year 9 | Year 10 |
|---|---|---|---|---|---|---|---|---|---|
| **AAA** | 375.232223 | 238.202248 | 316.508837 | 347.008533 | 329.046733 | 353.908703 | 301.266381 | 226.743239 | 293.729411 |
| **AA** | 449.595086 | 433.068045 | 302.396646 | 376.689919 | 320.197114 | 334.716658 | 401.673608 | 322.515591 | 440.834059 |
| **A** | 392.333146 | 378.611129 | 322.435595 | 324.567023 | 337.274540 | 351.092973 | 254.444730 | 352.720946 | 343.093259 |
| **BBB** | 400.255201 | 441.847670 | 490.490473 | 495.342060 | 265.000135 | 339.641702 | 349.681782 | 203.142461 | 320.187769 |
| **BB** | 356.204379 | 316.934964 | 464.740846 | 374.065129 | 387.404075 | 430.342011 | 353.291202 | 320.412204 | 409.923947 |
| **B** | 363.016749 | 377.035742 | 411.139123 | 277.599689 | 330.416669 | 248.657678 | 440.847600 | 308.988725 | 455.667702 |
| **C** | 352.386417 | 441.198342 | 324.206029 | 390.604030 | 315.164425 | 283.714896 | 329.104583 | 383.165025 | 364.021401 |
| **D** | 428.849730 | 455.866558 | 311.365330 | 319.144712 | 399.335395 | 268.793678 | 364.323453 | 178.505937 | 258.490771 |

| Year | Portfolio_EL_₹Cr |
|---|---|
| **Year 2** | 3117.872930 |
| **Year 3** | 3082.764698 |
| **Year 4** | 2943.282878 |
| **Year 5** | 2905.021094 |
| **Year 6** | 2683.839087 |
| **Year 7** | 2610.868299 |
| **Year 8** | 2794.633337 |
| **Year 9** | 2296.194127 |
| **Year 10** | 2885.948320 |

Interpretation:

- Expected Loss shows a gradual decline over the years, indicating an improving overall loss profile as exposures transition and stabilize across the rating spectrum.

## 2.2. Cumulative Default Probability (PD) Estimation Using Multi-Year Transition Matrix Approach

The average 1-year transition matrix was constructed based on year-to-year borrower rating movements and then row-normalised and adjusted to keep D as an absorbing state. For Year 1, EAD was calculated as Amount*EAD multiplier. From this, rating-wise EAD weights were derived. Multi-year cumulative PDs for each initial rating were obtained by using matrix powers: $T^5$ and $T^{10}$. Portfolio-level cumulative PDs were computed as the EAD-weighted average of rating-level cumulative PDs and provide a forward-looking indication of the expected default risk over 5- and 10-year horizons.

```
Portfolio 5-Year Cumulative PD (EAD-weighted at Year 1): 55.5134%
Portfolio 10-Year Cumulative PD (EAD-weighted at Year 1): 77.0662%
```

5-Year Cumulative PD by initial rating:

| | D |
|---|---|
| AAA | 47.4675% |
| AA | 48.6305% |
| A | 48.2914% |
| BBB | 48.4883% |
| BB | 49.0082% |
| B | 48.8573% |
| C | 48.4113% |
| D | 100.0000% |

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10-Year Cumulative PD by initial rating:

| | D |
|---|---|
| AAA | 72.9183% |
| AA | 73.5179% |
| A | 73.3431% |
| BBB | 73.4445% |
| BB | 73.7126% |
| B | 73.6348% |
| C | 73.4049% |
| D | 100.0000% |

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Interpretation:

- Cumulative PDs are high across all ratings, which indicates weak credit stability and strong downward migration.
- 5-year portfolio PD of 55.5% suggests more than half the exposure is expected to default within five years.
- 10-year portfolio PD of 77.1% points to severe long-term vulnerability with most exposures drifting toward default.
- There is minimal difference between high-grade and low-grade PDs, reflecting poor rating discrimination in the portfolio.
- The transition matrix demonstrates that the rating migrations are widely dispersed, rather than being concentrated near the diagonal, which indicates weak rating stability.
- Together with uniform EAD distribution across rating buckets, this suggests that no single rating dominates portfolio exposure, and the portfolio is broadly exposed to migration risk across all credit grades.

| to / from | AAA | AA | A | BBB | BB | B | C | D |
|---|---|---|---|---|---|---|---|---|
| AAA | 12.9916% | 12.6823% | 12.5055% | 13.2125% | 13.7870% | 12.0636% | 12.0194% | 10.7380% |
| AA | 13.0841% | 12.5501% | 12.1050% | 11.5710% | 11.8380% | 13.3066% | 12.8171% | 12.7281% |
| A | 12.4833% | 11.8145% | 12.7954% | 12.5279% | 13.7762% | 12.7954% | 11.6808% | 12.1266% |
| BBB | 13.1544% | 12.8025% | 12.8465% | 12.8905% | 11.3946% | 12.9784% | 11.4386% | 12.4945% |
| BB | 12.3269% | 13.3095% | 11.2550% | 12.2376% | 12.1483% | 12.3269% | 13.0415% | 13.3542% |
| B | 13.2100% | 11.2285% | 12.1532% | 13.0339% | 12.6376% | 11.8890% | 12.7257% | 13.1220% |
| C | 12.1239% | 12.9322% | 13.6057% | 11.7647% | 11.9892% | 12.1239% | 13.1118% | 12.3485% |
| D | 0.0000% | 0.0000% | 0.0000% | 0.0000% | 0.0000% | 0.0000% | 0.0000% | 100.0000% |

Start-of-period (Year 1) EAD by Rating and weights:

| Year 1 | ead_y1 | ead_w |
|---|---|---|
| AAA | 567418.3293% | 12.3648% |
| AA | 562518.6708% | 12.2580% |
| A | 560159.4610% | 12.2066% |
| BBB | 596195.5607% | 12.9919% |
| BB | 567389.9729% | 12.3641% |
| B | 592238.3267% | 12.9056% |
| C | 514596.7084% | 11.2137% |
| D | 628477.0626% | 13.6953% |

## 3. Stress Testing Framework: LGD Shocks and Rating Migration Impact Analysis

### 3.1. Impact of LGD Stress Factors on Year-Wise Expected Loss (Normal vs. Stressed EL)

The stressed scenario results in a systematic uplift in Expected Loss across all years, with increases typically in the 20-40% range. The impact is most pronounced in Years 3, 4, 7 and 9, signalling periods where the portfolio is structurally more exposed to LGD shocks. Overall, the pattern indicates that the portfolio exhibits moderate but persistent sensitivity to stress conditions, with stressed LGDs driving a meaningful escalation in annual loss outcomes.

```
Comparison of Portfolio EL (Normal vs. Stressed LGD):
--------------------------------------------------------------------------------
Year     | Normal EL (Rs Cr)      | Stress EL (Rs Cr)      | Increase (Rs Cr)
--------------------------------------------------------------------------------
2        | 3117.872929691799      | 4016.0577              |         +898.1848
3        | 3082.7646983223226     | 3983.5619              |         +900.7972
4        | 2943.282878010529      | 3901.2390              |         +957.9561
5        | 2905.021094416165      | 4029.6900              |        +1124.6689
6        | 2683.839087158584      | 3266.7607              |         +582.9216
7        | 2610.868298922203      | 3080.8059              |         +469.9376
8        | 2794.633337160309      | 3881.9775              |        +1087.3442
9        | 2296.1941267644893     | 3263.0906              |         +966.8965
10       | 2885.948319826173      | 3584.1569              |         +698.2086
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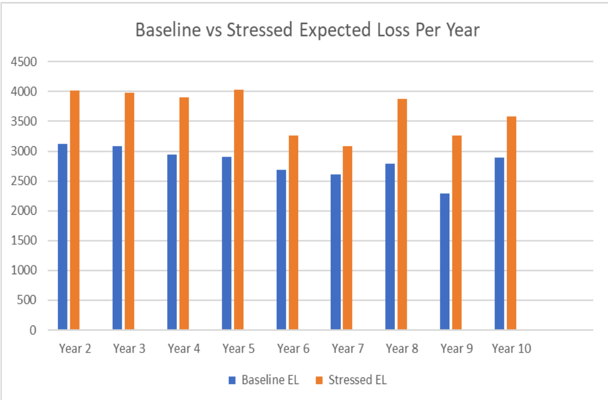
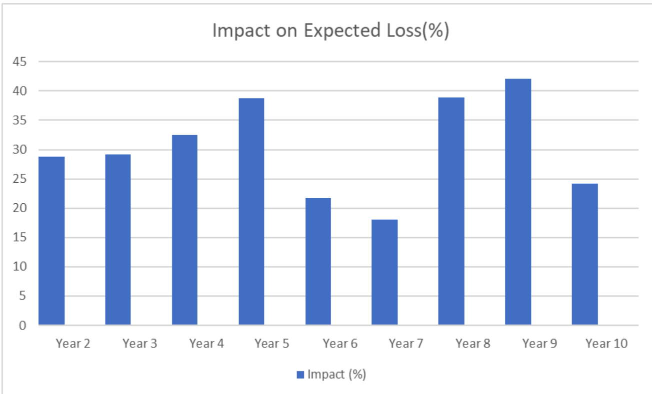
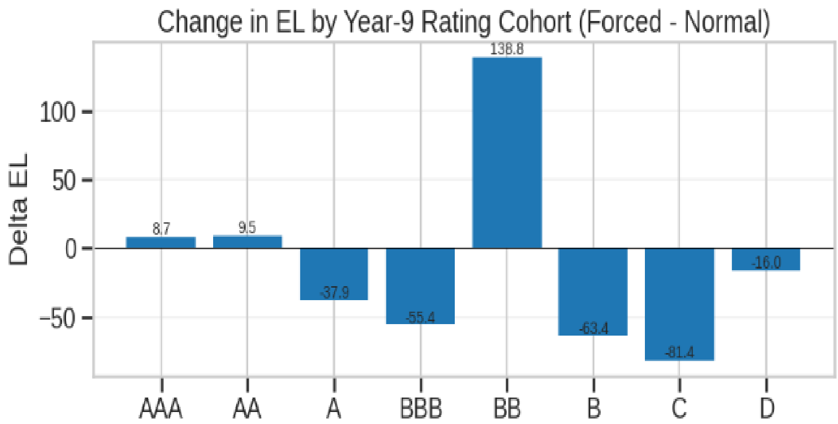



### 3.2. Impact of LGD Stress on Expected Loss Across Rating Categories

| Rating_Y9 | EL_Normal | EL_Stressed |
|---|---|---|
| AAA | 394.8025 | 450.4751 |
| AA | 387.0524 | 396.5948 |
| A | 360.5560 | 322.6739 |
| BBB | 331.3898 | 275.9487 |
| BB | 382.8938 | 521.7264 |
| B | 358.3002 | 294.8966 |
| C | 457.9477 | 376.5555 |
| D | 287.3415 | 271.3391 |




```
Portfolio Year-10 EL (Normal):   2960.2838
Portfolio Year-10 EL (Forced):   2910.2100
Difference (Forced - Normal):    -50.0738
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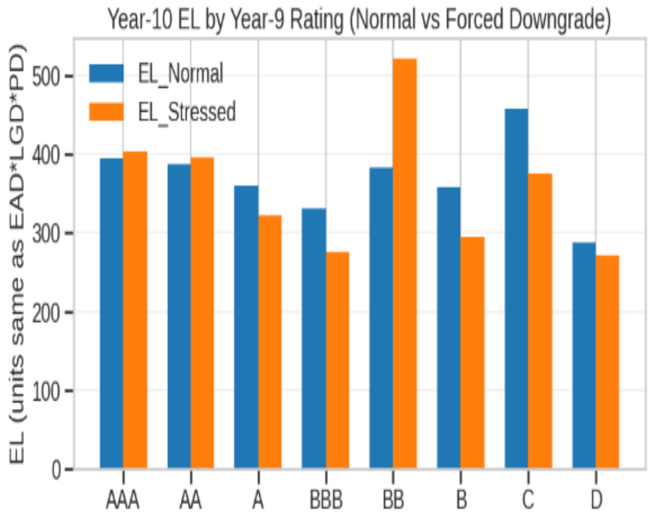
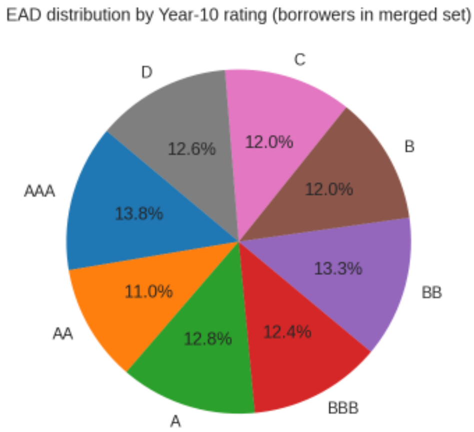



Interpretation:

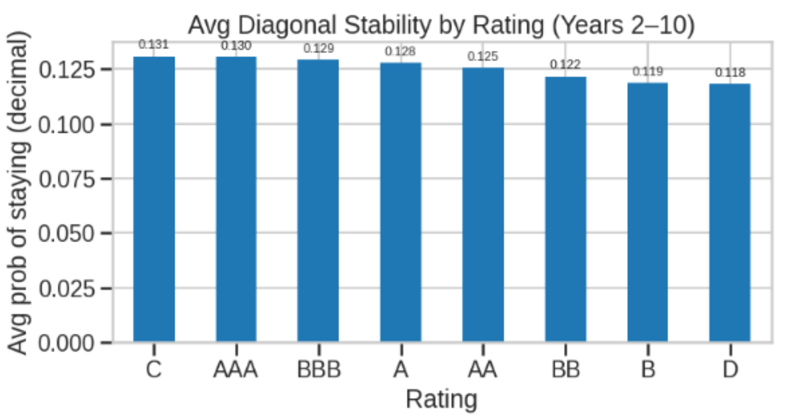
- The forced-downgrade scenario leads to a slightly lower portfolio EL (-₹50 Cr) compared to the actual Year-10 outcome.
- The reduction occurs because the weakest buckets (B, C, D) do not experience meaningful PD increases under the stress mapping, leading to lower ELs for these cohorts.
- Although the BB segment shows a sharp rise in losses, this is more than offset by reductions in other cohorts.
- Overall, the portfolio demonstrates limited sensitivity to the forced downgrade, with the net impact being marginally favourable rather than adverse.

## 4. Credit Migration Behaviour - Most Stable Rating

### 4.1. Rating Stability Methodology

Rating stability is measured using the diagonal elements of each year's transition matrix, which represent the probability that borrowers remain in the same rating from one year to the next. For each rating bucket, these stay probabilities are collected for all transitions between Year 2 to Year 10 and then averaged to obtain a long-term stability score.

| | Avg_Stability |
|---|---|
| C | 0.130792 |
| AAA | 0.130475 |
| BBB | 0.128998 |
| A | 0.127790 |
| AA | 0.125430 |
| BB | 0.121532 |
| B | 0.118674 |
| D | 0.118066 |




### 4.2. Rating Cohort Stability Analysis

It follows that the average diagonal probabilities across all grades are low, indicating frequent rating movements year to year. Overall, the portfolio demonstrates rather limited rating stickiness, thereby reinforcing the need to monitor migration risk closely. According to the average diagonal transition probability over Years 2–10, category C has the highest stability, which means that borrowers rated C tend to remain in the same category more consistently over time. Consequently, C is the most stable rating bucket in the portfolio.

## 5. Rating Trend & Migration Risk Indicators

### 5.1. Rating Trend & Migration Risk Indicators - Methodology

**Average Rating Trend** Each rating is converted into a numerical score (AAA=1,…,D=7). The average score for each year is then computed to track whether the portfolio is shifting toward better or worse credit quality.

$$\text{Avg Rating}_y = \frac{1}{N}\sum_{i=1}^{N} \text{Score}(R_{i,y})$$

**Portfolio-Level PD Trend** For each year, borrower-level PDs are mapped from the rating-specific PD vector, and the portfolio PD is taken as the simple average across borrowers. This gives a yearly view of expected default risk based on observed ratings.

$$\text{Portfolio PD}_y = \frac{1}{N}\sum_{i=1}^{N} PD(R_{i,y})$$

**Downgrade Ratio**

The downgrade ratio measures the proportion of borrowers whose rating worsens between Year y and y+1. Higher values indicate stronger downward migration pressure.

$$\text{Downgrade Ratio}_{y \to y+1} = \frac{\#\{R_{i,y+1} > R_{i,y}\}}{N}$$

### 5.2. Identifying Significant Shifts in Portfolio Risk

A significant shift in portfolio risk can be identified by comparing the year-on-year rating distributions using a statistical test such as the Kolmogorov–Smirnov (KS) test. The KS statistic measures how different two distributions are, and the p-value indicates whether this difference is statistically meaningful.

- If p-value $< 0.05 \rightarrow$ the shift is significant.
- If p-value $> 0.05 \rightarrow$ no statistically significant change.

Example from the portfolio

For the transition Year 3 → Year 4, the KS statistics are:

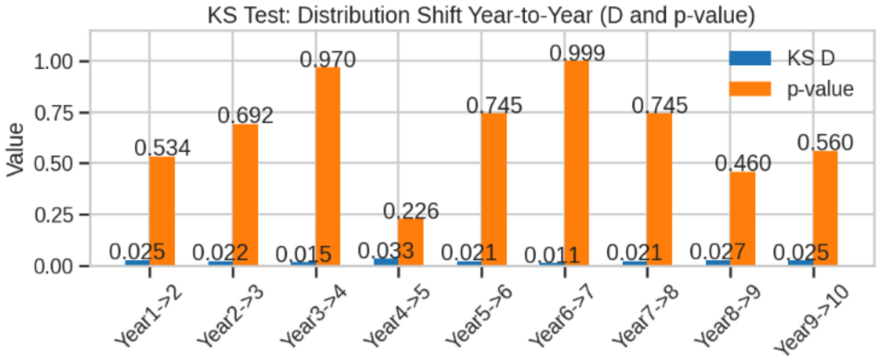
- KS_D = 0.0155
- p-value = 0.9699

Since the p-value is much greater than 0.05, we conclude that the rating distribution in Year 4 is not significantly different from Year 3. This means the portfolio's credit-risk profile has not materially shifted, even though some borrowers may have upgraded or downgraded individually.

Interpretation:

- This approach separates normal migration noise from true structural risk change. Only when a year pair shows large KS_D and low p-value would we flag a meaningful deterioration or improvement in portfolio credit quality.
- In this portfolio, KS_D values are very small (0.01–0.03) and all p-values are comfortably above 0.05, meaning none of the year pairs show statistically significant shifts. This confirms that, despite some downgrade activity (around 42–50% each year), the overall shape of the rating distribution has remained stable.

- The downgrade ratio stays consistently high at ~43–50% each year, indicating steady downward rating movement. This shows that nearly half the portfolio is downgraded annually, reflecting persistent credit weakening rather than a one-off shock.

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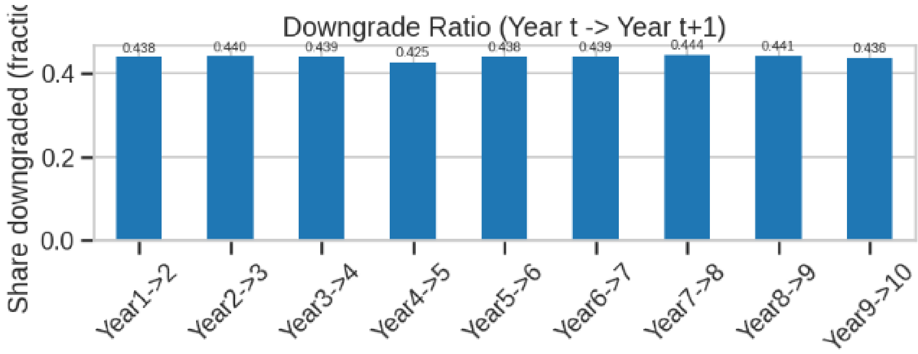


## 6. Conclusion

While diversified across ratings, the credit portfolio of the bank demonstrates high migration volatility with low rating stability. The annual ELs remain stable, while the long-term cumulative PDs are very high, indicating structural credit weakness and poor rating discrimination. Stress tests indicated moderate sensitivity, particularly in the mid-tier buckets like BB, which drive most of the loss amplification. Without statistically significant year-on-year shifts, the portfolio has shown a persistent drift toward weaker credit quality, reflecting heightened long-term credit risks and the need to operate under tighter risk controls and sharper rating governance.